MIL-STD-1640B
7 February 1985
SUPERSEDING
MIL-STD-1640A
1 JUNE 1979

MILITARY STANDARD

MIXER STAGES, RADIO FREQUENCY, SELECTION OF

FSC 5985

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DEPARTMENT OF DEFENSE
WASHINGTON, DC 20363

Mixer Stages, Radio Frequency, Selection of

MIL-STD-1640B

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ELEX 8111, Department of the Navy, Washington, DC 20363 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1. SCOPE

1.1 Scope. This standard provides a list of mixer stages for use in military equipment applications.

1.2 Purpose of standard. The purpose of this standard is for the following:

a. Provide new equipment designers with a list of mixer stages considered for use in military applications.

b. Restrict the number of mixer stages for use in military applications in order to provide effective logistic support of equipment.

c. Establish criteria pertinent to choice and application of mixer stages for use in military equipment.

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2. REFERENCED DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications, standards, and handbooks, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-28837 - Mixer Stages, Radio Frequency, General Specification For.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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3. DEFINITIONS

3.1 The terms used in this standard are those commonly encountered in mixer stages engineering practice.

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4. GENERAL REQUIREMENTS

4.1 Selection of mixer stages. Mixer stages to be used in military applications shall be selected from those listed in tables I through VIII.

4.2 Criteria for selection. The criteria for the selection of mixer stages for inclusion in this standard are as follows:

a. The mixer stages shall be considered by representatives of the Military Departments the best available type for current application.

b. Availability of the mixer stages shall be reasonably certain.

c. The mixer stages shall have an approved military specification.

4.3 Application and use. Mixer stages used in military applications shall be representative of manufactured lots possessing acceptable material, and physical and electrical characteristics. They shall in no manner degrade the operational characteristics of the equipment it is used in.

4.4 Detailed requirements for mixer stages. The detailed requirements for mixer stages listed in this standard are covered by the applicable MIL-M-28837 specification sheet.

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5. DETAILED REQUIREMENTS

Not applicable.

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6. NOTES

6.1 Changes from previous issue. Asterisks or vertical lines are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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TABLE I. Mixer stages with 8-pin dual inline headers.

Part number M28837/1-	Frequency range (MHz)	LO power input (dBm)	Maximum conversion loss (SSB) (dB) (MHz)	Noise figure (SSB) (dB) (MHz)	ISOLATION (SSB) minimum LO-RF dB	LO-IF dB	RF-IF dB	LO frequency (MHz)
01	LO .05 - 200 RF .05 - 200 IF DC - 200	+7	6.5 at f_L and f_R 0.2 - 50, f_I DC - 50	6.5 at f_L and f_R 1 - 50, f_I 0.4 - 50	45	40	N/A	.05 - 30
			8.0 at f_L and f_R 50 - 200, f_I DC - 200	8.0 at f_L and f_R 50 - 200, f_I 0.4 - 200	35	30	N/A	30 - 200
			8.5 at f_L and f_R 0.05 - 0.2, f_I DC - 0.4					
02	LO 5 - 500 RF 5 - 500 IF DC - 500	+7	7.0 at f_L and f_R 5 - 150, f_I 0.4 - 150	7.0 at f_L and f_R 10 - 100, f_I 0.4 - 100	45	40	N/A	5 - 50
			9.0 at f_L and f_R 150 - 500, f_I 0.4 - 500	9.0 at f and f 100 - 500, f_I 0.4 - 500	30	25	N/A	50 - 500
03	LO .5 - 500 RF .5 - 500 IF DC - 500	+7	7.5 at f_L and f_R 1 - 250, f_I DC - 250	N/A	35	30	N/A	0.5 - 1
			8.5 at f_L and f_R 0.5 - 500, f_I DC - 500		30	25	N/A	1 - 250
					25	20	N/A	250 - 500
04	LO 2 - 500	+20	6.5 at f_I 5 - 80, 7.0 at f_I .4 - 120; f_L 5 - 200, f_R 20 - 80	6.5 at f_I 5 - 80, 7.0 at f_I DC - 120; f_L 5 - 200, f_R 20 - 80	40	40	N/A	2 - 32
	RF 2 - 400		7.0 at f_I 5 - 80, 8.0 at f_I DC - 300; f_L 5 - 300, f_R 5 - 300	7.0 at f_I 5 - 80, 8.0 at f_I 2 - 120; f_L 5 - 300, f_R 5 - 300	35	35	N/A	32 - 100
	IF DC - 800		7.5 at f_I 5 - 80; f_L 5 - 470, f_R 5 - 400	8.0 at f_I 5 - 80; f_L 5 - 470, f_R 5 - 400	25	25	N/A	100 - 500
			9.0 at f_I DC - 800; f_L 2 - 500, f_R 2 - 400	9.5 at f_I 2 - 800; f_L 2 - 500, f_R 2 - 400	N/A	N/A	20	2 - 400
05	LO .05 - 200 RF .05 - 200 IF DC - 200	+13	7.5 at f_L and f_R 0.2 - 50, f_I DC - 50	7.5 at f_L and f_R 1 - 50, f_I 0.4 - 50	45	40	N/A	.05 - 30
			9.0 at f_L and f_R .05 - 200, f_I DC - 200	9.0 at f_L and f_R 50 - 200, f_I 0.4 - 200	30	25	N/A	30 - 200
					N/A	N/A	30	.05 - 3

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TABLE I. Mixer stages with 8-pin dual inline headers - Continued.

Part number M28837/1-	Frequency range (MHz)	LO power input (dBm)	Maximum conversion loss (SSB) (dB) (MHz)	Noise figure (SSB) (dB) (MHz)	ISOLATION (SSB) minimum LO-RF dB	LO-IF dB	RF-IF dB	LO frequency (MHz)
06	LO .5 - 500 RF .5 - 500 IF DC - 500	+17	7.0 at f_L and f_R 0.5 - 30, f_I DC - 30	7.0 at f_L and f_R 1 - 30, f_I 0.4 - 30	55	45	N/A	0.5 - 30
			7.5 at f_L and f_R 30 - 100, f_I DC - 100	7.5 at f_L and f_R 30 - 100, f_I 0.4 - 100	45	35	N/A	30 - 100
					35	25	N/A	100 - 500
			9.0 at f_L and f_R 100 - 500, f_I DC - 500	9.0 at f_L and f_R 100 - 500, f_I 0.4 - 500	N/A	N/A	20	0.5 - 500
07	LO 5 - 400 RF 5 - 400 IF DC - 400	+17	7.5 at f_L and f_R 5 - 200, f_I DC - 200	N/A	35	30	N/A	5 - 100
			9.0 at f_L and f_R 200 - 400, f_I DC - 400		25	20	N/A	100 - 400
08	LO 0.5 - 500 RF 0.5 - 500 IF DC - 500	+17	7.5 at f_L and f_R 0.5 - 5	N/A	45	35	N/A	0.5 - 5
			8.5 at f_L and f_R 5 - 500, f_I DC - 500		30	30	N/A	5 - 250
					25	20	N/A	250 - 500
09	LO 1 - 750 RF 1 - 750 IF DC - 750	+7	7.5 at f_L and f_R 2 - 375, f_I DC - 375	N/A	45	30	N/A	1 - 2
			8.5 at f_L and f_R 1 - 750, f_I DC - 750		30	25	N/A	2 - 375
					25	20	N/A	375 - 750
10	LO .4 - 500 RF .4 - 500 IF DC - 500	+13	7.5 at f_L and f_R 1 - 50, f_I 0.4 - 50	7.5 at f_L and f_R 1 - 50, f_I 0.4 - 50	45	40	N/A	0.4 - 50
			9.0 at f_L and f_R 0.4 - 500, f_I 0.4 - 500	9.0 at f_L and f_R 50 - 500, f_I 0.4 - 500	25	25	N/A	50 - 500
11	LO 1 - 500 RF 1 - 500 IF DC - 500	+23	8.5 at f_L and f_R 2 - 400, f_I DC - 400	Within 1 dB of conversion loss	50 40 30 20	40 30 20 20	25 20 20 20	1 - 100 100 - 200 200 - 300 300 - 500
			9.5 at f_L and f_R 1 - 500, f_I DC - 500					
12	LO .002 - 12 RF .002 - 12 IF DC - 12	+7	6.0 at f_L and f_R .01 - 5, f_I 0.4 - 5	6.0 at f_L and f_R .01 - 5, f_I 0.4 - 5	45	40	N/A	.002 - 5
			8.0 at f_L and f_R .002 - 12, f_I 0.4 - 12	8.0 at f_L and f_R 5 - 12, f_I 0.4 - 12	40	30	N/A	5 - 12

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TABLE I. Mixer stages with 8-pin dual inline headers - Continued.

Part number M28837/1-	Frequency range (MHz)	LO power input (dBm)	Maximum conversion loss (SSB) (dB) (MHz)	Noise figure (SSB) (dB) (MHz)	ISOLATION (SSB) minimum LO-RF dB	LO-IF dB	RF-IF dB	LO frequency (MHz)
13	LO .05 - 200	0	6.5 at f_L and f_R 0.2 - 50, f_I DC - 50	6.5 at f_L and f_R 0.2 - 50, f_I 0.4 - 50	45	40	N/A	.05 - 30
	RF .05 - 200		8.0 at f_L and f_R 50 - 200, f_I DC - 200	8.0 at f_L and f_R 50 - 200, f_I 0.4 - 200	35	30	N/A	30 - 200
	IF DC - 200		8.5 at f_L and f_R 0.05 - 0.2, f_I DC - 0.2					
14	LO 5 - 500	0	7.0 at f_L and f_R 10 - 100, f_I 10 - 100	7.0 at f_L and f_R 10 - 100, f_I 10 - 100	45	40	N/A	5 - 50
	RF 5 - 500		8.0 at f_L and f_R 100 - 200, f_I 10 - 200	8.0 at f_L and f_R 100 - 200, f_I 10 - 200	30	25	N/A	50 - 500
	IF DC - 500		9.5 at f_L and f_R 5 - 500, f_I .5 - 500	9.5 at f_L and f_R 5 - 500, f_I .5 - 500				
15	LO 1 - 500, RF 1 - 500	+7	7.0 at f_L and f_R 1 - 400, f_I DC - 400	Within 1 dB of conversion loss	40	35	N/A	1 - 100
	IF DC - 500		8.0 at f_L and f_R 1 - 500, f_I DC - 500		30	25	N/A	100 - 500
16	LO 1 - 500, RF 1 - 500	+15	7.0 at f_L and f_R 1 - 400, f_I DC - 400	Within 1 dB of conversion loss	40	40	N/A	1 - 100
	IF DC - 500		8.0 at f_L and f_R 400 - 500, f_I DC - 500		30	25	N/A	100 - 500

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TABLE II. Mixer stages with solder leads, flat-pack configuration.

Part number M28837/2-	Frequency range (MHz)	LO power input (dBm)	Maximum conversion loss (SSB) (dB) (MHz)	Noise figure (SSB) (dB) (MHz)	ISOLATION (SSB) minimum LO-RF dB	LO-IF dB	RF-IF dB	LO frequency (MHz)
01	RF 5 - 1,000 LO 5 - 1,000 IF DC - 1,000	+10	8 at f_L and f_R 5 - 1,000 f_I DC - 1,000	Within 1 dB of conversion loss	40	40	30	5 - 100
					30	25	15	100 - 1,000
02	RF 10 - 1,500	+7	7.0 at f_R 20 - 600 f_L 10 - 800 f_I DC - 200	7.0 at f_R 20 - 600 f_L 10 - 800 f_I 0.4 - 200	30	30	N/A	10 - 600
	LO 10 - 1,500		8.0 at f_R 10 - 1,200 f_L 10 - 1,400 f_I DC - 200	8.0 at f_R 10 - 1,200 f_L 10 - 1,400 f_I 0.4 - 200	25	20	N/A	600 - 1,200
	IF DC - 1,000		f_R 10 - 1,500 f_L 10 - 1,500 8.5 at f_I DC - 200 9.5 at f_I DC - 1,000	f_R 10 - 1,500 f_L 10 - 1,500 8.5 at f_I 0.4 - 200 9.5 at f_I 0.4 - 1,000	25	18	N/A	1,200 - 1,500
03	RF 600 - 2,000 LO 600 - 2,000	+7	9.0 at f_L and f_R 600 - 1,000 f_I DC - 1,000	Within 1 dB of conversion loss	25	23	25	600 - 1,000
	IF DC - 1,000		7.5 at f_L and f_R 1,000 - 2,000 f_I DC - 1,000		20	12	15	1,000 - 2,000
04	RF .5 - 500 LO .5 - 500	+7	7.0 at f_L and f_R 1 - 300 f_I DC - 300	Within 1 dB of conversion loss	40	30	23	0.5 - 300
	IF DC - 500		8.0 at f_L and f_R 0.5 - 500 f_I DC - 500		35	20	20	0.5 - 500
05	RF .5 - 500 LO .5 - 500 IF_1 DC - 500 IF_2 DC - 500	+7	7.0 at RF port to IF_2 .5 - 500 7.0 at LO port to IF_1 .5 - 500	Within 1 dB of conversion loss	35 30 25	30 25 20	25 20 15	0.5 - 10 10 - 200 200 - 500
06	RF .5 - 400 LO .5 - 600	+17	6.5 at f_L and f_R 1 - 100 f_I DC - 100	6.5 at f_L and f_R 1 - 100 f_I DC - 100	45	30	20	1 - 100
	IF DC - 400		8.0 at f_L and f_R .5 - 400 f_I DC - 400	8.0 at f_L and f_R .5 - 400 f_I DC - 400	35	25	10	.5 - 400

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TABLE II. Mixer stages with solder leads, flat-pack configuration - Continued.

Part number M28837/2-	Frequency range (MHz)	LO power input (dBm)	Maximum conversion loss (SSB) (dB) (MHz)	Noise figure (SSB) (dB) (MHz)	ISOLATION (SSB) minimum LO-RF dB	LO-IF dB	RF-IF dB	LO frequency (MHz)
07	RF 500 - 1,000 LO 500 - 1,000 IF DC - 1,000	+7	7.5 at f_L and f_R 5 - 500, f_I DC - 500	Within 1 dB of conversion loss	35	30	25	5 - 50
					30	25	20	50 - 500
			8.0 at f_L and f_R 500 - 1,000, f_I DC - 1,000		25	20	15	500 - 1,000
08	RF 750 - 2,000 LO 500 - 2,000 IF DC - 1,200	0	7.0 at f_L and f_R 500 - 1,000, f_I DC - 1,000	Within 1 dB of conversion loss	35	27	21	500 - 1,000
			10 at f_L and f_R 1,000 - 2,000, f_I DC - 1,200		27	24	17	1,000 - 2,000
09	RF 1 - 3,500 LO 1 - 3,500 IF 5 - 2,500	+10	7.0 at f_L and f_R 5 - 1,000, f_I DC - 1,000	Within 1 dB of conversion loss	30	30	30	5 - 1,000
			9.5 at f_L and f_R 1 - 3,500, f_I DC - 2,500		20	20	18	1 - 3,500
10	RF 800 - 2,400 LO 800 - 3,500 IF DC - 1,500	+7	8.0 at f_R 1,000 - 2,000, f_L 800 - 3,500, f_I 10 - 1,500	8.0 at f_R 1,000 - 2,000, f_L 800 - 3,500, f_I 10 - 1,500	25	18	20	800 - 2,000
			8.5 at f_R 800 - 2,400, f_L 800 - 3,500, f_I 10 - 1,500	8.5 at f_R 800 - 2,400, f_L 800 - 3,500, f_I 10 - 1,500	20	20	20	2,000 - 3,500
11	RF 2.5 - 5.5 GHz LO 2.5 - 7.0 GHz IF DC - 1.5 GHz	+9	6.5 at f_L 3 - 5.5 GHz, f_R 3 - 5.0 GHz, f_I 0.03 - 0.5 GHz	6.5 at f_L 3 - 5.5 GHz, f_R 3 - 5.0 GHz, f_I 0.03 - 0.5 GHz	30			2.5 - 7 GHz
						17		2.5 - 3.5GHz
							20	3.5 - 7 GHz
			7.0 at f_L 2.5 - 7 GHz, f_R 2.5 - 5.5 GHz, f_I 0.03 - 1.5 GHz	7.0 at f_L 2.5 - 7 GHz, f_R 2.5 - 5.5 GHz, f_I 0.03 - 1.5 GHz				
12	RF 4.5 - 9.5 GHz LO 2.5 - 11.5 GHz IF DC - 2.0 GHz	+10	7.0 at f_R 5 - 9 GHz, f_L 4 - 10 GHz, f_I 0.03 - 1 GHz	7.0 at f_R 5 - 9 GHz, f_L 4 - 10 GHz, f_I 0.03 - 1 GHz	25			2.5 - 9 GHz
					20			9 - 11.5 GHz
						15		4 - 11.5 GHz
						10		2.5 - 4 GHz
			8.0 at f_R 4.5 - 9.5 GHz, f_L 2.5 - 11 GHz, f_I 0.03 - 2 GHz	8.0 at f_L 4.5 - 9.5 GHz, f_R 2.5 - 11 GHz, f_I 0.03 - 2 GHz			15	4.5 - 8.0 GHz
							18	8 - 9.5 GHz

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TABLE II. Mixer stages with solder leads, flat-pack configuration - Continued.

Part number M28837/2-	Frequency range (GHz)	LO power input (dBm)	Maximum conversion loss (SSB) (dB) (GHz)	Noise figure (SSB) (dB) (GHz)	ISOLATION (SSB) minimum LO-RF dB	LO-IF dB	RF-IF dB	LO frequency (GHz)
13	RF 7 - 18 LO 5 - 18 IF DC - 3	+10	8.0 at f_R 8 - 16, f_L 5 - 18, f_I 0.03 - 3	8.0 at f_R 8 - 16, f_L 5 - 18, f_I 0.03 - 3	22			5 - 14
					15			14 - 18
						12		5 - 8
						22		8 - 18
			8.5 at f_R 8 - 16, f_L 5 - 16, f_I 0.03 - 3	8.5 at f_R 8 - 16, f_L 5 - 16, f_I 0.03 - 3			23	0.03 - 8
							15	8 - 18
			9.0 at f_R 16 - 18, f_L 13 - 18, f_I 0.03 - 3	f_R 16 - 18, f_L 13 - 18, f_I 0.03 - 3				
14	RF 1 - 18 LO 2 - 18 IF DC - 5	+13	8.0 at f_R 5 - 13, f_L 5 - 13, f_I 0.03 - 2	8.0 at f_R 5 - 13, f_L 5 - 13, f_I 0.03 - 2	18	20		2 - 18
							25	1 - 2
							28	2 - 8
			9.0 at f_R 2 - 16, f_L 2 - 18, f_I 0.03 - 4	9.0 at f_R 2 - 16, f_L 2 - 18, f_I 0.03 - 4				
			10.0 at f_R 1 - 18, f_L 2 - 18, f_I 0.03 - 5	10.0 at f_R 1 - 18, f_L 2 - 18, f_I DC - 5				
15	RF 2 - 18 LO 2 - 18 IF 1 - 8	+13	10.0 at f_R 2 - 10, f_L 2 - 18, f_I 1 - 18	10.0 at f_R 2 - 10, f_L 2 - 18, f_I 1 - 18	15	16	20	2 - 18
			10.5 at f_R 10 - 18, f_L 10 - 18, f_I 2 - 8	10.5 at f_R 10 - 18, f_L 10 - 18, f_I 2 - 8				
			11.0 at f_R 10 - 18, f_L 2 - 10, f_I 1 - 8	11.0 at f_R 10 - 18, f_L 2 - 10, f_I 1 - 8				
16	RF 2 - 18 LO 2 - 18 IF DC - 4.0	+10	10 at f_R 2 - 10, f_L 2 - 14, f_I 0.03 - 4	10 at f_R 2 - 10, f_L 2 - 14, f_I 0.03 - 4	15	16	20	2 - 8
							15	8 - 18
			11 at f_R 10 - 18, f_L 6 - 18, f_I 0.03 - 4	11 at f_R 10 - 18, f_L 6 - 18, f_I 0.03 - 4				

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TABLE III. Mixer stages with SMA connectors (female).

Part number	Frequency range (MHz)	LO power input (dBm)	Maximum conversion loss (SSB) (dB) (MHz)	Noise figure (SSB) (dB) (MHz)	ISOLATION (SSB) minimum LO-RF dB	LO-IF dB	RF-IF dB	LO frequency (MHz)
M28837/3-01	RF 8,000 - 12,400 LO 8,000 - 12,400 IF DC - 1,000	+7	f_L and f_R 8,000 - 12,400 7.5 at f_I DC - 200 10.5 at f_I 200 - 1,000	f_L and f_R 8,000 - 12,400 7.5 at f_I DC - 200 11.5 at f_I 200 - 1,000	-6	N/A	N/A	8,000 - 12,400
M28837/5-01	RF 300 - 1,000 LO 300 - 1,000 IF DC - 1,000	+7	8.0 at f_L and f_R 300 - 1,000 f_I DC - 200	8.0 at f_L and f_R 300 - 1,000 f_I 10 - 200	40	25	20	300 - 1,000
			9.0 at f_I 200 - 1,000	9.0 at f_I 200 - 1,000				
M28837/5-02	RF 1,000 - 4,200 LO 1,000 - 4,200 IF DC - 1,000	+7	8.5 at f_L and f_R 1,000 - 1,500 f_I DC - 1,000	8.5 at f_L and f_R 1,000 - 1,500 f_I 30 - 1,000	25	15	N/A	1,000 - 4,200
			7.5 at f_L and f_R 1,500 - 4,200 f_I DC - 1,000	7.5 at f_L and f_R 1,500 - 4,200 f_I 30 - 1,000				
M28837/5-03	RF 0.2 - 500 LO 0.2 - 500	+7	6.5 at f_L, f_R and f_I 1 - 50	6.5 at f_L, f_R and f_I 1 - 50	45	40	35	2 - 50
	IF DC - 500		8.5 at f_L, f_R and f_I 2 - 500	8.5 at f_L, f_R and f_I 1 - 500	35	25	20	50 - 500
M28837/5-04	RF 1,000 - 4,000 LO 1,000 - 4,000 IF DC - 1,000	+20	8.0 at f_L 1,200 - 4,000 f_R 1,200 - 3,500 f_I DC - 500	8.0 at f_L 1,200 - 4,000 f_R 1,200 - 3,500 f_I DC - 500	20		N/A	1,000 - 4,000
						10	"	1,000 - 2,000
						15	"	2,000 - 4,000
			9.5 at f_L and f_R 1,000 - 4,000 f_I DC - 1,000	9.5 at f_L and f_R 1,000 - 4,000 f_I DC - 1,000				
M28837/5-05	RF 2,500 - 5,500 LO 2,500 - 7,000 IF DC - 1,500	+9	6.5 at f_L 3,000 - 5,500 f_R 3,000 - 5,000 f_I 30 - 500	6.5 at f_L 3,000 - 5,500 f_R 3,000 - 5,000 f_I 30 - 500	30			2,500 - 7,000
						17		2,500 - 3,500
							20	3,500 - 7,000
			7.0 at f_L 2,500 - 7,000 f_R 2,500 - 5,500 f_I 30 - 1,500	7.0 at f_L 2,500 - 7,000 f_R 2,500 - 5,500 f_I 30 - 1,500				
M28837/5-06	RF 4,500 - 9,500 LO 2,500 - 11,500 IF DC - 2,000	+10	7.0 at f_L 4,000 - 10,000 f_R 5,000 - 9,000 f_I 30 - 1,000	7.0 at f_L 4,000 - 10,000 f_R 5,000 - 9,000 f_I 30 - 1,000	25			2,500 - 9,000
					20			9,000 - 11,500
						15		4,000 - 11,500
						10		2,500 - 4,000
			8.0 at f_L 2,500 - 11,000 f_R 4,500 - 9,500 f_I 30 - 2,000	8.0 at f_L 2,500 - 11,000 f_R 4,500 - 9,500 f_I 30 - 2,000			15	4,500 - 8,000
							18	8,000 - 9,500

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TABLE III. Mixer stages with SMA connectors (female) - Continued.

Part number	Frequency range (MHz)	LO power input (dBm)	Maximum conversion loss (SSB) (dB) (MHz)	Noise figure (SSB) (dB) (MHz)	ISOLATION (SSB) minimum LO-RF dB	LO-IF dB	RF-IF dB	LO frequency (MHz)
M28837/5-07	RF 7,000 - 18,000 LO 5,000 - 18,000 IF DC - 3,000	+10	8.0 at f_L 5,000 - 18,000 f_R 7,000 - 16,000 f_I 30 - 3,000	8.0 at f_L 5,000 - 18,000 f_R 7,000 - 16,000 f_I 30 - 3,000	22			5,000 - 14,000
					15			14,000 - 18,000
			8.5 at f_L 5,000 - 16,000 f_R 8,000 - 16,000 f_I 30 - 3,000	8.5 at f_L 5,000 - 16,000 f_R 8,000 - 16,000 f_I 30 - 3,000		12		5,000 - 8,000
						22		8,000 - 18,000
			9.0 at f_L 13,000 - 18,000 f_R 16,000 - 18,000 f_I 30 - 3,000	9.0 at f_L 13,000 - 18,000 f_R 16,000 - 18,000 f_I 30 - 3,000			23	30 - 8,000
							15	8,000 - 18,000
M28837/5-08	RF 1,000 - 18,000 LO 2,000 - 18,000 IF DC - 5,000	+13	8.0 at f_L 5,000 - 13,000 f_R 5,000 - 13,000 f_I 30 - 2,000	8.0 at f_L 5,000 - 13,000 f_R 5,000 - 13,000 f_I 30 - 2,000	18	20		2,000 - 18,000
							25	1,000 - 2,000
			9.0 at f_L 2,000 - 18,000 f_R 2,000 - 16,000 f_I 30 - 4,000	9.0 at f_L 2,000 - 18,000 f_R 2,000 - 16,000 f_I 30 - 4,000			28	2,000 - 8,000
			10.0 at f_L 2,000 - 18,000 f_R 1,000 - 18,000 f_I 30 - 5,000	10.0 at f_L 2,000 - 18,000 f_R 1,000 - 18,000 f_I DC - 5,000				
M28837/5-09	RF 2,000 - 18,000 LO 2,000 - 18,000 IF DC 1,000 - 8,000	+13	10.0 at f_L 2,000 - 18,000 f_R 2,000 - 10,000 f_I 1,000 - 8,000	10.0 at f_L 2,000 - 18,000 f_R 2,000 - 10,000 f_I 1,000 - 8,000	15	16	20	2,000 - 18,000
			10.5 at f_L 10,000 - 18,000 f_R 10,000 - 18,000 f_I 2,000 - 8,000	10.5 at f_L 10,000 - 18,000 f_R 10,000 - 18,000 f_I 2,000 - 8,000				
			11.0 at f_L 2,000 - 10,000 f_R 10,000 - 18,000 f_I 2,000 - 8,000	11.0 at f_L 2,000 - 10,000 f_R 10,000 - 18,000 f_I 2,000 - 8,000				
M28837/5-10	RF 2,000 - 18,000 LO 2,000 - 18,000 IF DC - 4,000	+10	10.0 at f_L 2,000 - 14,000 f_R 2,000 - 10,000 f_I 30 - 4,000	10.0 at f_L 2,000 - 14,000 f_R 2,000 - 10,000 f_I 30 - 4,000	15	16		2,000 - 18,000
							20	2,000 - 8,000
			11.0 at f_L 6,000 - 18,000 f_R 10,000 - 18,000 f_I 30 - 4,000	11.0 at f_L 6,000 - 18,000 f_R 10,000 - 18,000 f_I 30 - 4,000			15	8,000 - 18,000

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TABLE IV. Mixer stages with Series N and BNC connectors (female).

Part number	Frequency range	IF frequency	Input power
	GHz	MHz	mW
M28837/4-01	4-8	30	1-2

TABLE V. Mixer stages with BNC connectors (female).

Part number	Frequency range (MHz)	LO power input (dBm)	Maximum conversion loss (SSB) (dB) (MHz)	Noise figure (SSB) (dB) (MHz)	ISOLATION (SSB) minimum LO-RF dB	LO-IF dB	RF-IF dB	LO frequency (MHz)
M28837/6-01	LO 3.0 - 1,000 RF 3.0 - 1,000 IF DC - 1,000	+7	7.5 at f_L and f_R 10 - 100, f_I DC - 100	7.5 at f_L and f_R 10 - 100, f_I 0.4 - 100	40	40	N/A	3 - 100
			10 at f_L and f_R 3 - 1,000, f_I DC - 1,000	10 at f_L and f_R 3 - 1,000, f_I 0.4 - 1,000	30	20	N/A	100 - 1,000

TABLE VI. Mixer stages with TO-5 configuration.

Part number	Frequency range (MHz)	LO power input (dBm)	Maximum conversion loss (SSB) (dB) (MHz)	Noise figure (SSB) (dB) (MHz)	ISOLATION (SSB) minimum LO-RF dB	LO-IF dB	RF-IF dB	LO frequency (MHz)
M28837/7-01	RF 10 - 500 LO 10 - 500 IF DC - 500	+7	7.0 at f_L and f_R 10 - 200, f_I DC - 200	7.0 at f_L and f_R 10 - 200, f_I 0.4 - 200	40 35	35 30	25 20	10 - 50 50 - 100
			8.0 at f_L and f_R 200 - 350, f_I DC - 350	8.0 at f_L and f_R 200 - 350, f_I 0.4 - 350	30 25	25 15	15 10	100 - 200 200 - 500
			9.0 at f_L and f_R 350 - 500, f_I DC - 500	9.0 at f_L and f_R 350 - 500, f_I 0.4 - 500				
M28837/7-03	RF 0.4 - 500 LO 0.4 - 500 IF DC - 500	+7	6.5 at f_L and f_R 1 - 200, f_I DC - 200	6.5 at f_L and f_R 2 - 200, f_I 2 - 200	45	30	N/A	0.4 - 60
			7.5 at f_L and f_R 0.4 - 500, f_I DC - 500	7.5 at f_L and f_R 2 - 500, f_I 2 - 500	30	20	N/A	60 - 500

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TABLE VII. Mixer stages with PC mount configurations.

Part number	Frequency range (MHz)	LO power input (dBm)	Maximum conversion loss (SSB) (dB) (MHz)	Noise figure (SSB) (dB) (MHz)	ISOLATION (SSB) minimum LO-RF dB	LO-IF dB	RF-IF dB	LO frequency (MHz)
M28837/8-01	RF DC - 800 LO 5 - 800 IF DC - 800	+7	6.5 at f_L and f_R 5 - 500, f_I DC - 500	6.5 at f_L and f_R 5 - 500, f_I 0.4 - 500	40	40	N/A	5 - 50
					30	30	N/A	50 - 100
			7.0 at f_L and f_R 500 - 800, f_I DC - 800	7.0 at f_L and f_R 500 - 800, f_I 0.4 - 800	20	20	N/A	100 - 800

TABLE VIII. Mixer stages with TO configurations.

Part number	Frequency range (MHz)	LO power input (dBm)	Maximum conversion loss (SSB) (dB) (MHz)	Noise figure (SSB) (dB) (MHz)	ISOLATION (SSB) minimum LO-RF dB	LO-IF dB	RF-IF dB	LO frequency (MHz)
M28837/9-01	RF 10 - 1,000 LO 10 - 1,000 IF DC - 1,200	0	8.5 at f_L and f_R 10 - 1,000, f_I DC - 1,200	Within 1 dB of conversion loss	40	35	30	10 - 50
					20	25	15	50 - 750
					15	17	10	750 - 1,000
M28837/9-02	RF 10 - 1,000 LO 10 - 1,000 IF DC - 600	+20	7.5 at f_L and f_R 10 - 300, f_I 10 - 200	7.5 at f_L and f_R 10 - 300, f_I 10 - 200	35	35		10 - 100
					25	25		100 - 400
			9.0 at f_L and f_R 10 - 300, f_I 1 - 600	9.0 at f_L and f_R 10 - 300, f_I 2 - 600	18	18		400 - 1,000
							12	10 - 1,000
			10.0 at f_L and f_R 10 - 1,000, f_I 1 - 500	10.0 at f_L and f_R 10 - 1,000, f_I 2 - 500				
M28837/9-03	RF 10 - 1,500 LO 10 - 1,500 IF DC - 600	+7	7.2 at f_R 200 - 600, f_L 10 - 800, f_I 1 - 200	7.2 at f_R 20 - 600, f_L 10 - 800, f_I 1 - 200	35	30		10 - 500
					28	20		500 - 1,200
					25	18		1,200 - 1,500
			9.0 at f_L and f_R 10 - 1,500, f_I 1 - 800	9.0 at f_L and f_R 10 - 1,500, f_I 1 - 800				
M28837/9-04	RF 10 - 1,600 LO 10 - 1,600 IF DC - 800	+13	7.5 at f_L 10 - 800, f_R 20 - 600, f_I 0.4 - 200	7.5 at f_L 10 - 800, f_R 20 - 600, f_I 0.4 - 200	35	28		10 - 500
					28	20		500 - 1,000
			8.5 at f_L and f_R 10 - 1,600, f_I 0.4 - 800	8.5 at f_L and f_R 10 - 1,600, f_I 0.4 - 800	25	15		1,000 - 1,600

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